

5	(a)	(i)	$R_R = 105\Omega$ $R_{Th} = 50\Omega$	1 1
		(ii)	$V = \frac{50}{105+50} \times 12 = 3.87\text{ V}$	1 1
		(iii)	The p.d across resistor and thermistor is equal so this implies that the resistance across each component is equal. Hence the temperature is 50 °C	1 1
	(b)	(i)	p.d across length $x = \frac{x}{l} E$ Since the appliance is parallel to length of x, p.d is the same. Thus p.d across the appliance is $x = \frac{x}{l} E$	1 1
		(ii)	p.d across appliance $= 20 / 100 \times 50 = 10\text{ V}$	1